Machine learning assignment 1
Turtle age regression

Course: Introduction to Machine Learning (822047-B-6) Spring 2025

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1. Introduction

Non-invasive age prediction using machine learning to predict turtle age based on physical characteristics is of great importance, because of the nature of the researched group. Traditionally, to predict the age of a turtle, the shell would be sliced to count the growth rings (Germano and Bury, 1998). This is, especially with an endangered species like turtles, highly unethical. Moreover, it is very impractical for large-scale studies. This project will therefore explore different models to estimate the number of growth rings using the turtle's features.

2. Data and Preprocessing

The dataset consists of information about 4177 turtles. This data includes continuous variables: length, diameter, height, whole weight, shucked weight, viscera weight, shell weight, and a categorical variable sex with values 'M' (Male), 'F' (Female), and 'I' (infant). The target variable is 'Rings', which represents the number of growth rings on a turtle's shell and serves as an indirect indicator of its age. One-hot encoding was applied to the categorical variable to make it suitable for modeling.

Before the data could be splitted, the continuous target was binned where several binning strategies were evaluated. With 3 bins, class imbalance was evident: Bin-1 contained 45.29%, Bin-2 34.63%, and Bin-0 only 20.08%. Increasing to 4 bins, Bin-3 held 34.66%, while others ranged from 13.66% to 27.6%. With 5 bins, the distribution improved: Bin-3 contained 26.85%, Bin-4 22.99%, Bin-1 22.96%, Bin-2 16.49%, and Bin-0 10.72%. When increasing the bins further, two bins would contain most of the percentage, while others contained around 10%. Thus, the 5-bin strategy was selected to preserve distributional balance across the train (80%), validation (10%), and test (10%) sets.

3. Model Training and Tuning

Three regression models were evaluated: Linear Regression, Polynomial Regression, and K-Nearest Neighbors (KNN). Linear Regression served as a baseline. Polynomial models were evaluated for degrees 2 to 5. KNN was evaluated for k {1, 3, 6, 10}. Models were trained on the training set and validated on the 10% validation set. The best hyperparameters were selected based on validation R^2 score.

Linear Regression performed reasonably with R^2 = 0.501 on the test set, but had the highest MSE (0.882) and MAE (0.743), indicating less precise predictions. Polynomial Regression improved performance, with degree 2 resulting in the best balance between complexity and accuracy. Higher degrees caused instability and overfitting, leading to extremely negative R2 values. KNN performed well at k=10, benefiting from averaging predictions over similar neighboring instances. While both Polynomial Regression (degree 2) and KNN (k=10) showed similar R^2 scores on the test set 0.5577 and 0.5533 respectively Polynomial was chosen as the final model. This decision was based on three factors: slightly higher R^2, better interpretability through feature coefficients, and its compatibility with coefficient-based analysis tasks such as identifying most influential features. Although KNN had a better MAE (0.6785 vs. 0.6830), Polynomial's overall balance and interpretability made it more appropriate for detailed analysis. (Figure 1)

4. Results and Discussion

A 10-fold cross-validation was performed using the polynomial model (Figure 2). The R^2 scores across folds were stable with a mean near 0.575, suggesting the model generalizes moderately well. Coefficient analysis revealed the most influential features were weight-related, including Whole Weight and its interaction with Sex (Figure 3), which was formed in the polynomial model.

Error analysis was conducted on the worst-performing fold. Over-predicted turtles were typically long and wide but light in weight, causing the model to infer greater age than true. Under-predicted cases had shorter shells, leading the model to systematically underestimate their age. This indicates that shell length may be misleading without context from other physical attributes such as weight. (Figure 4 & 5)

5. Conclusion

This project showed that machine learning, particularly KNN and Polynomial Regression, can be effective non-invasive methods for estimating turtle age using their features. The most effective model was Polynomial Regression (degree 2), which offered strong predictive performance and interpretability. Its coefficients enabled feature ranking and deepened understanding of age predictors. The model demonstrated systematic error biases related to shell size and weight interactions which will be helpful to consider for future work. Besides, future work could incorporate more biological or environmental variables, larger datasets, or ensemble methods to further improve prediction accuracy.

Appendix

Figure 1:

model	Hyperparameters	R^2	MSE	MAE
Linear regression	NA	0.5008829767688672	0.8820388813049345	0.7425648857164858
Polynomial regression	Degree = 2	0.5577373078570338	0.781565989665804	0.6830495259408175
KNN regression	K=10	0.5533438265127667	0.7893301435406699	0.6784688995215311

Figure 2:

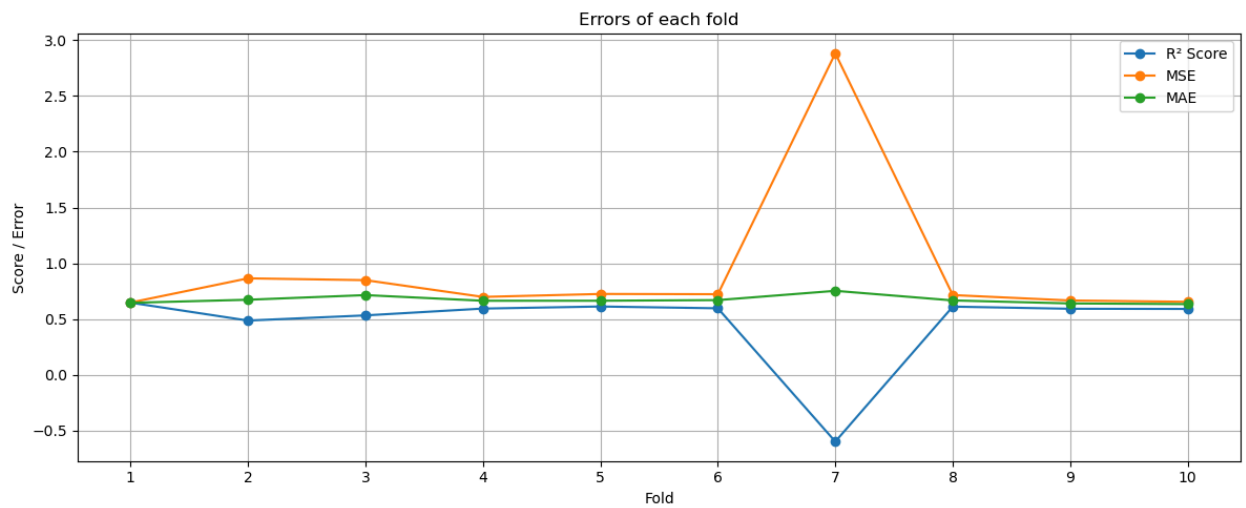


Figure 3:

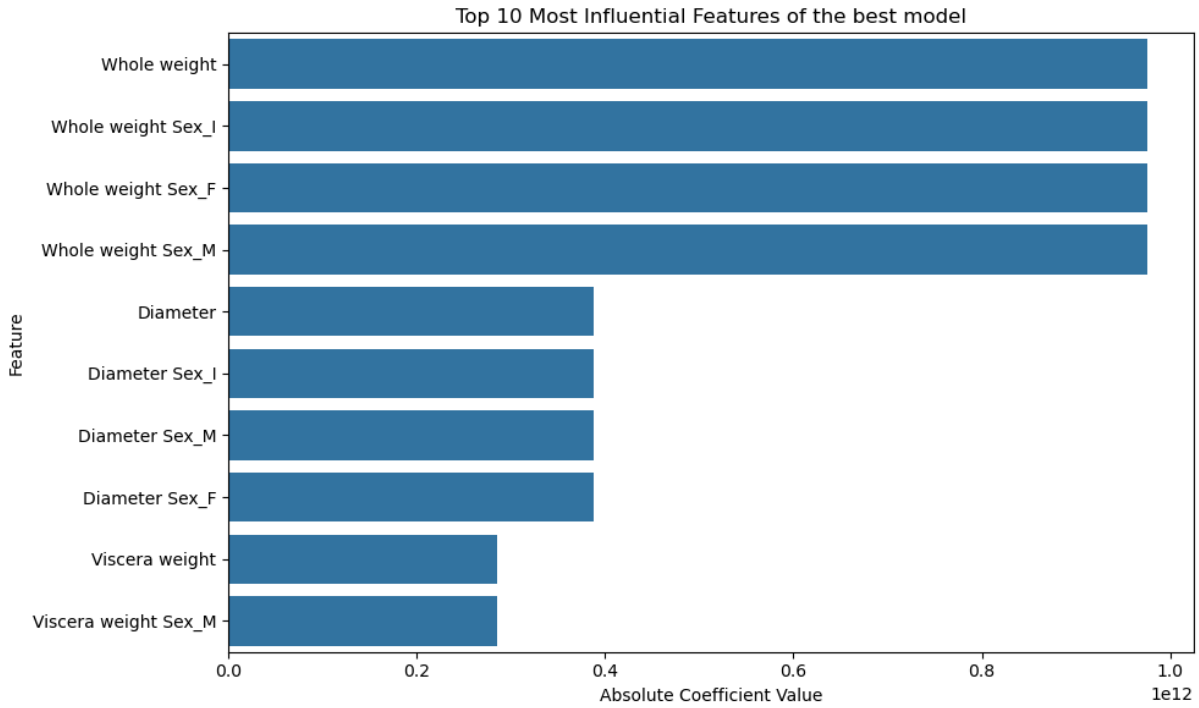


Figure 4:

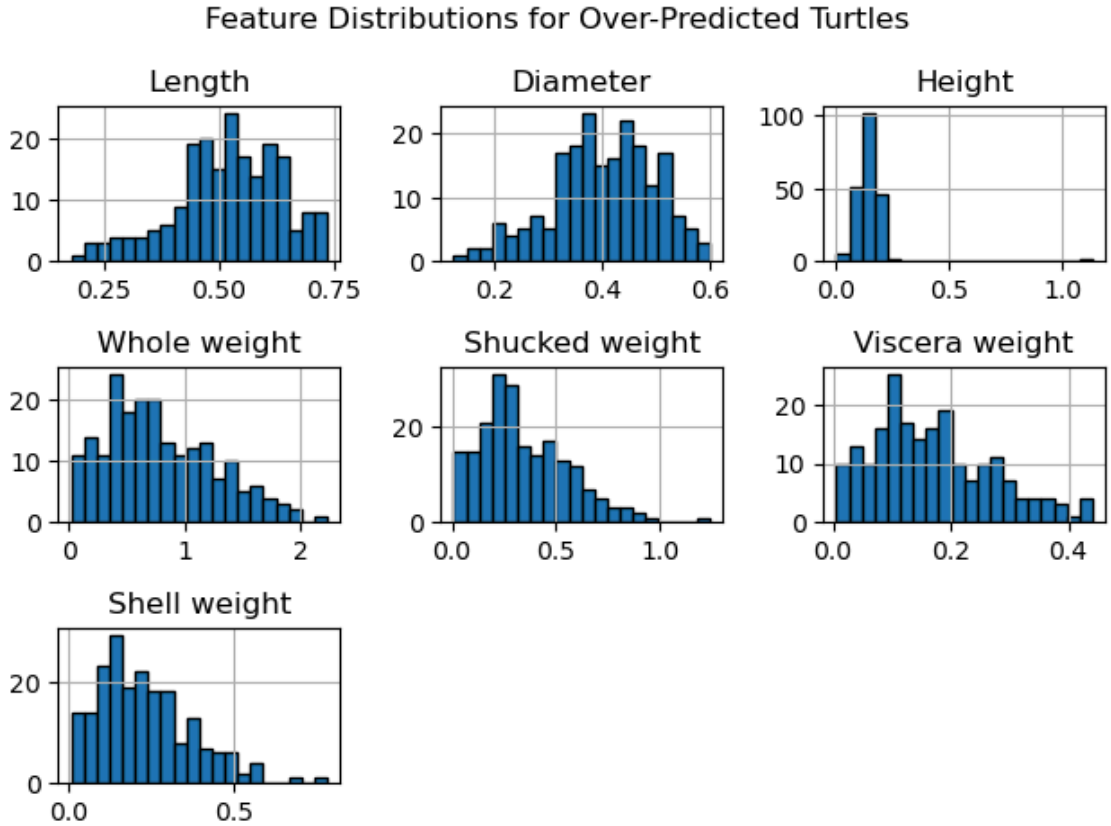
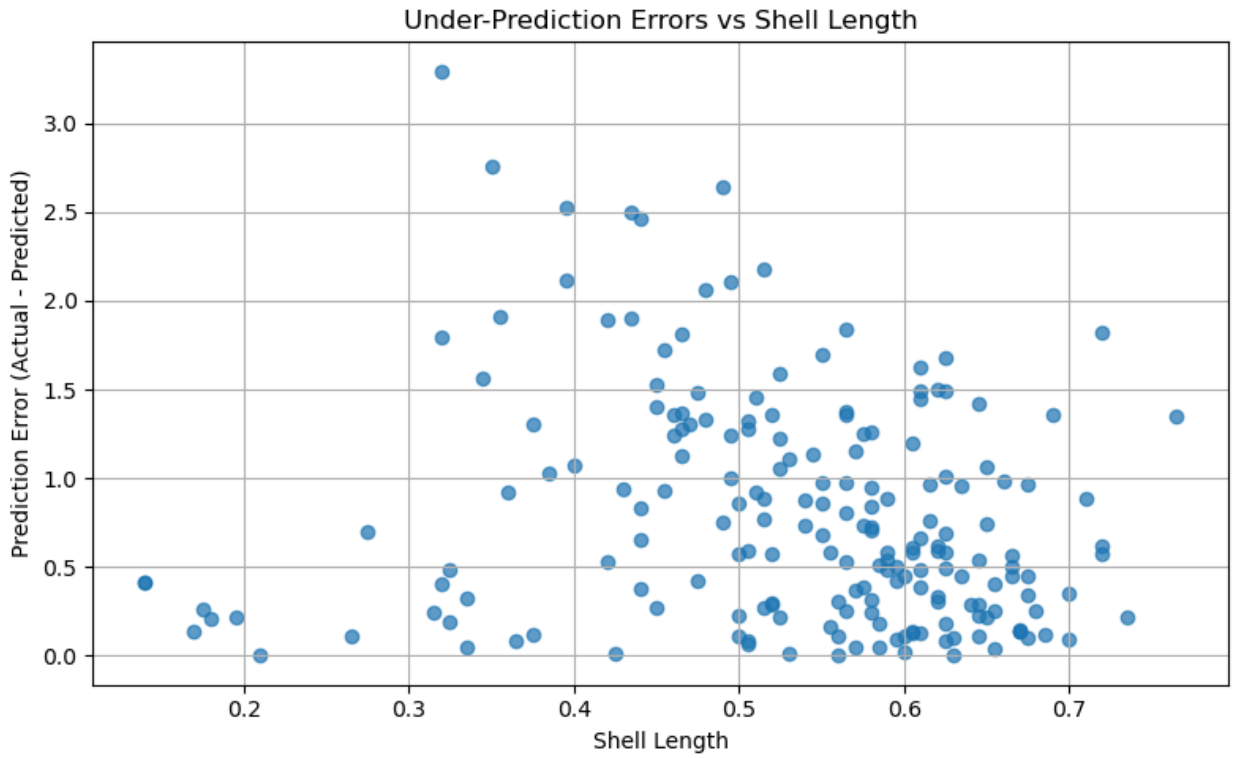


Figure 5:


Under-Prediction Errors vs Shell Length
Prediction Error (Actual - Predicted)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0.2
0.3
0.4
0.5
0.6
0.7
Shell Length

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